

MULTIPLE COMPARISONS IN A MIXED TWO-WAY MODEL

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Key words: Dunnett's method; intraclass correlation model; mixed two-way model; multiple comparison; Scheffé's method; Tukey's method.

Summary: This article presents an alternative and simpler proof of the results of Bhargava and Srivastava (1973) and of Hochberg and Tamhane (1983) on the validity of the methods of multiple comparisons (Tukey, Dunnett and Scheffé) in the mixed two-way model. This paper gives a shorter and easier alternative proof of those results, which may also be applied to more general designs.

1. INTRODUCTION

Graybill (1961, chapter 18) sets out a mixed two-way model in which each observation $x_{ijh} = \mu + \alpha_i + b_j + c_{ij} + e_{ijh}$ is the sum of a general mean (μ), plus the fixed effect (α_i) of each row, plus the random effect (b_j) of each column, plus the random interaction (c_{ij}), and of the random error (e_{ijh}), with $i = 1, 2, \dots, r$; $j = 1, 2, \dots, t$; $h = 1, 2, \dots, n$. In this model the variables $e_{ijh} \sim N(0, \sigma^2)$, $b_j \sim N(0; \sigma_B^2)$ and $c_{ij} \sim N(0, (r-1)\sigma_{AB}^2/r)$ are all independent from each other for any index with the exception of $\text{Cov}(c_{ij}, c_{i'j}) = -\alpha_{AB}^2/r$. Then it follows that the mean vector of fixed treatments $(\bar{x}_{1..}, \bar{x}_{2..}, \dots, \bar{x}_{r..})$ is distributed as an r -dimensional normal distribution with means $E(\bar{x}_{i..}) = \mu + \alpha_i$, variances $V(\bar{x}_{i..}) = \{r\sigma^2 + nr\sigma_B^2 + n(r-1)\sigma_{AB}^2\} \div rtn$, and covariances $\text{Cov}(\bar{x}_{i..}, \bar{x}_{i'..}) = (r\sigma_B^2 - \sigma_{AB}^2)/rt = C$. This is equivalent to what Scheffé (1956) called the symmetric model, and it is known in the

literature as the intraclass correlation model. Bhargava and Srivastava (1973) proved that the classical methods of multiple comparisons apply to this model for the case $n = 1$; Hochberg and Tamhane (1983) extended the result to the case $n > 1$. This paper gives a shorter and easier alternative proof of those results, which may also be applied to more general designs.

2. THE BASIC THEOREM

THEOREM: Let $(x_1, x_2, \dots, x_k)$ be a random vector normally distributed with covariance matrix R given by $R_{ii} = V$ and $R_{ij} = C$ ($i \neq j$). Then, it is always possible to define a new vector $(y_1, y_2, \dots, y_k)$ which is normally distributed with $E(y_i) = E(x_i)$, $V(y_i) = V - C$ and $\text{Cov}(y_i, y_j) = 0$ ($i \neq j$).

PROOF: Let us consider $y_i = x_i + x$, where x is a normal random variable with $E(x) = 0$ and $V(x)$ and $\text{Cov}(x, x_i)$ have yet to be fixed. Then $E(y_i) = E(x_i)$, and:

(a) If $C < 0$, by taking $V(x) = -C$ and $\text{Cov}(x, x_i) = 0$, it follows that $V(y_i) = V - C$ and $\text{Cov}(y_i, y_j) = C - C = 0$ ($i \neq j$).

(b) If $C > 0$, by taking $V(x) = C$ and $\text{Cov}(x, x_i) = -C$, it follows that $V(y_i) = V - 2C + C = V - C$ and $\text{Cov}(y_i, y_j) = C - 2C + C = 0$.

Note that to choose $V(x) = C$ and $\text{Cov}(x, x_i) = -C$ is possible provided that the covariance matrix, R^+ , of the vector $(x_1, x_2, \dots, x_k, x)$ is positive definite. Indeed, since R is positive definite, we need only to verify that the determinant of the matrix R^+ is positive. But

$$\det \begin{bmatrix} V & C & \dots & C & -C \\ C & V & \dots & C & -C \\ \cdot & \cdot & \dots & \cdot & \cdot \\ C & C & \dots & V & -C \\ -C & -C & \dots & -C & C \end{bmatrix} = \det \begin{bmatrix} V-C & 0 & \dots & 0 & 0 \\ 0 & V-C & \dots & 0 & 0 \\ \cdot & \cdot & \dots & \cdot & \cdot \\ 0 & 0 & \dots & V-C & 0 \\ -C & -C & \dots & -C & C \end{bmatrix} = (V-C)^k C > 0$$

and the proof follows.

The theorem has the practical advantage of allowing the application of the classical methods of multiple comparisons to the x_i ($i \geq 1$) values, by means of the y_i ($i \geq 1$) ones, which are already independent, provided that one has an appropriate estimator for $V - C$, which is χ^2 distributed. The theorem also applies to non-normal variables.

3. MULTIPLE COMPARISONS IN A MIXED TWO-WAY MODEL

By applying the theorem to the variables $x_i = \bar{x}_{i..}$, we can conclude that there exist normal independent variables y_i with means $E(y_i) = \mu + \alpha_i$ and variances $V(y_i) = V - C = (\sigma^2 + n\sigma_{AB}^2)/nt$. Moreover the mean squared error s_{AB}^2 of interaction of the mixed two-way model is such that $fs_{AB}^2 \sim (\sigma^2 + n\sigma_{AB}^2) \chi_f^2$, where $f = (r-1)(t-1)$, so that s_{AB}^2 is an unbiased estimator of the unknown variance $\{\sigma^2 + n\sigma_{AB}^2\}$. It is also independent of the random variables y_i , which follows from the fact that it is independent of the x_i . Multiple comparisons results for the mixed two-way model are easily derived from our theorem:

COROLLARY 1: The $\begin{bmatrix} r \\ 2 \end{bmatrix}$ differences $\alpha_i - \alpha_j$ verify simultaneously, with probability $1 - \alpha$, that

$$(1) \quad \alpha_i - \alpha_j \in (\bar{x}_{i..} - \bar{x}_{j..}) \pm q_{r,f}^{\alpha} \{s_{AB}^2/nt\}^{\frac{1}{2}}$$

where $q_{r,f}$ is the studentized range.

PROOF: One need only apply the result in section 3.6 of Scheffé (1959) to the y_i random variables. ■

COROLLARY 2: The $(r-1)$ differences $\alpha_i - \alpha_1$ ($i \neq 1$) verify simultaneously, with probability $1 - \alpha$, that

$$(2) \quad \alpha_i - \alpha_1 \in (\bar{x}_{1..} - \bar{x}_{i..}) \pm d_{r-1,f}^{\alpha} \{2s_{AB}^2/nt\}^{\frac{1}{2}}$$

where $d_{r-1,f}^{\alpha}$ is the Dunnett distribution.

PROOF: One need only apply the result in section 2.5 of Miller (1966) to the y_i random variables. ■

COROLLARY 3: The set of all the possible contrasts for the form $\sum \lambda_i \alpha_i$ ($\sum \lambda_i = 0$) verify simultaneously, with probability of $1 - \alpha$, that

$$(3) \quad \sum \lambda_i \alpha_i \in \sum \lambda_i \bar{x}_{i..} \pm J_{r,f}^{\alpha} \{s_{AB}^2 (\sum \lambda_i^2)/nt\}^{\frac{1}{2}}$$

where $J_{r,f}^{\alpha} = \{(r-1)F_{r-1,f}^{\alpha}\}^{\frac{1}{2}}$ and $F_{r-1,f}$ is the Snedecor F distribution with the same degrees of freedom.

PROOF: One need only apply the result of section 3.4 of Scheffé (1959) to the y_i random variables. ■

From expressions (1), (2) and (3), and from the fact that $V(\bar{x}_{1..} - \bar{x}_{i..}) = 2(\sigma^2 + n\sigma_{AB}^2)/nt$, the conclusion can be drawn that the random variables $\bar{x}_{i..}$, in the case of multiple comparisons, may be treated as if they were independent with variance $(\sigma^2 + n\sigma_{AB}^2)/nt$, or equivalently, that the random variables x_{ijh} , in the same case, are independent with variance $\{\sigma^2 + n\sigma_{AB}^2\}$, which may be estimated by means of the mean squared error s_{AB}^2 of the row of interaction with $(r-1)(t-1)$ degrees of freedom.

These results are valid in general for every K -dimensional vector normally distributed with a matrix of covariances of type R indicated above.

ACKNOWLEDGMENTS

I am grateful to the referee for his useful comments.

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